PROBLEM 26

Heavy Particle on a Rotating Half-Circumference (Moving-Frame Method)

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Rotating Wire + Inertia Forces

A heavy point particle P of mass m is constrained to move without friction along the half-circumference wire of radius a shown in the figure. The wire containing the point particle rotates around the OY axis with constant angular velocity $\boldsymbol{\omega} = \Omega\mathbf{j}$. Initially, the particle is located at $x = 0$ with velocity v_0 along the positive x axis.

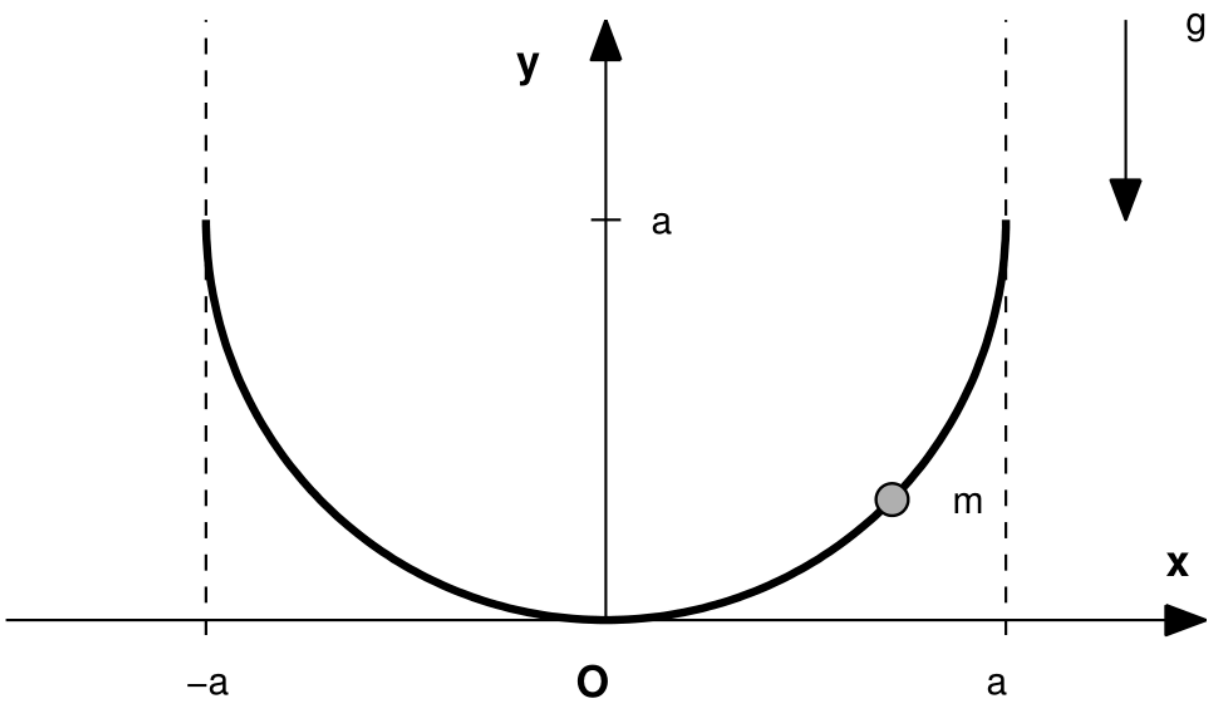


Figure: Half-circumference wire of radius a with lowest point O, ends at x = ±a, rotating about the vertical axis OY; particle of mass m on the wire.

(a) Determine the number of degrees of freedom of the problem.

(b) Define a convenient moving reference frame with origin in O, $OX_1Y_1Z_1$. Express the inertia force that acts on the particle in this moving reference frame.

(c) Express also the components of the weight of the particle with respect to $OX_1Y_1Z_1$.

(d) Determine if there are any equilibrium points.

(e) Reduce the problem to integrals (i.e. quadratures) and determine the value of the reaction force.

(f) Is it possible to use the energy equation? If Ω were a function of time, would it be possible to use the energy equation? Justify your answer.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical system:** a heavy particle P (mass m) slides without friction on a half-circumference wire of radius a. In the plane of the wire the lowest point is O, the centre is $C = (0, a)$ and the ends are $(\pm a, a)$ (figure). The wire turns about the vertical axis OY with constant angular velocity $\boldsymbol{\omega} = \Omega\,\mathbf{j}$.
- **Hypotheses:** the frame $OXYZ$ with OY vertical upwards is inertial; gravity is $-g\,\mathbf{j}$; the wire is smooth, so its reaction has no component along the wire; the rotation is imposed (whatever torque is needed is supplied). This problem was solved in Topic 3; here it is solved again with the relative-motion method of Topic 5 (moving frame, Notes Eq. 3.27 and inertia forces, Notes Eqs. 4.62–4.63).
- **Why a moving frame:** in a frame that turns with the wire the constraint is a fixed circle, so the motion of P is a one-dimensional motion on a fixed curve. The cost is to add the centrifugal and Coriolis forces; there is no Euler force and no force from the acceleration of the origin because Ω is constant and O is fixed.
- **Initial conditions ($t = 0$):** P at O ($x = 0$) with velocity v_0 along $+x$.

Symbol	Meaning	SI unit
m	mass of P	kg
a	radius of the wire	m
Ω	angular velocity of the wire (constant)	rad/s
g	gravitational acceleration	m/s^2
v_0	initial speed at O	m/s
θ	angle of CP from the downward vertical	rad

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial frame:** $S_0 = OXYZ = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$, with $\mathbf{j}$ upwards.
- **Moving frame** (requested in (b)): $S_1 = OX_1Y_1Z_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ attached to the wire, with OX_1 in the plane of the wire (the x axis of the figure), $OY_1 \equiv OY$ and OZ_1 normal to the plane of the wire. Taking S_1 coincident with S_0 at $t = 0$, it is a simple rotation of angle Ωt about $\mathbf{j}$ (theory note, Sec. 4.1):

$$\mathbf{i}_1 = \cos\Omega t\,\mathbf{i} - \sin\Omega t\,\mathbf{k}, \qquad \mathbf{j}_1 = \mathbf{j}, \qquad \mathbf{k}_1 = \sin\Omega t\,\mathbf{i} + \cos\Omega t\,\mathbf{k}$$

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & 0 & \sin\Omega t \\ 0 & 1 & 0 \\ -\sin\Omega t & 0 & \cos\Omega t \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\Omega t + \sin^2\Omega t = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

(the columns are unit vectors and $\mathbf{i}_1 \cdot \mathbf{k}_1 = \cos\Omega t \sin\Omega t - \sin\Omega t \cos\Omega t = 0$; also $\mathbf{i}_1 \times \mathbf{j}_1 = \mathbf{k}_1$).

- **Motion of S_1: $\mathbf{r}_0^O = \mathbf{0}$, $\mathbf{v}_0^O = \mathbf{0}$, $\mathbf{a}_0^O = \mathbf{0}$** (fixed origin); $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}_1$, $\boldsymbol{\alpha}_{10} = \dot{\Omega}\,\mathbf{j}_1 = \mathbf{0}$. Check: $\left.\frac{d\mathbf{i}_1}{dt}\right|_0 = -\Omega\sin\Omega t\,\mathbf{i} - \Omega\cos\Omega t\,\mathbf{k} = -\Omega\,\mathbf{k}_1$ (chain rule), and $\Omega\,\mathbf{j} \times \mathbf{i}_1 = -\Omega\,\mathbf{k}_1$.
- **Position of P in S_1:** with θ the angle of CP measured from the downward vertical, positive towards $+X_1$, and $\theta \in [-\pi/2, \pi/2]$ on the wire:

$$\mathbf{r}_1^P = a\sin\theta\,\mathbf{i}_1 + a\,(1 - \cos\theta)\,\mathbf{j}_1$$

- **Wire basis (intrinsic frame of the circle):** tangent $\mathbf{e}_t = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1$, principal normal towards C, $\mathbf{e}_n = -\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1$, binormal $\mathbf{e}_t \times \mathbf{e}_n = (\cos^2\theta + \sin^2\theta)\,\mathbf{k}_1 = \mathbf{k}_1$. The matrix with columns $\mathbf{e}_t, \mathbf{e}_n, \mathbf{k}_1$ in $\mathcal{B}_1$ is a rotation by θ about $\mathbf{k}_1$, with determinant $\cos^2\theta + \sin^2\theta = 1$.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Degrees of freedom

Pedagogical Justification: CDOF = (coordinates of the free particle) − (independent holonomic constraints).

P has 3 coordinates. In S_1 the wire imposes two fixed equations, $z_1 = 0$ (plane of the wire) and $x_1^2 + (y_1 - a)^2 = a^2$ (circle). In S_0 the same constraints are rheonomic (they turn with Ωt), but the count is the same: CDOF $= 3 - 2 = 1$, with generalized coordinate θ. The rotation Ωt is prescribed, so it is not a degree of freedom.

(b) Inertia force in $OX_1Y_1Z_1$

Pedagogical Justification: Notes Eq. 4.62 (Slide 28) gives $\mathbf{F}_i = -m\mathbf{a}_0^O - m\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P - m\boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P) - 2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$. The first two terms vanish ($\mathbf{a}_0^O = \mathbf{0}, \boldsymbol{\alpha}_{10} = \mathbf{0}$).

Relative velocity. In S_1 only the components change; chain rule $\frac{d}{dt}\sin\theta = \cos\theta\,\dot\theta$, $\frac{d}{dt}(1 - \cos\theta) = \sin\theta\,\dot\theta$:

$$\mathbf{v}_1^P = a\dot\theta\cos\theta\,\mathbf{i}_1 + a\dot\theta\sin\theta\,\mathbf{j}_1 = a\dot\theta\,\mathbf{e}_t$$

Centrifugal force. With $\mathbf{j}_1 \times \mathbf{i}_1 = -\mathbf{k}_1$ and $\mathbf{j}_1 \times \mathbf{j}_1 = \mathbf{0}$, $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = \Omega\,\mathbf{j}_1 \times a\sin\theta\,\mathbf{i}_1 = -\Omega a\sin\theta\,\mathbf{k}_1$; then, with $\mathbf{j}_1 \times \mathbf{k}_1 = \mathbf{i}_1$, $\boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P) = -\Omega^2 a\sin\theta\,\mathbf{i}_1$:

$$\mathbf{F}_C = m\Omega^2 a\sin\theta\,\mathbf{i}_1$$

It points horizontally away from the axis and equals $m\Omega^2\mathbf{r}_\perp$ (Slide 30), since $\mathbf{r}_\perp = a\sin\theta\,\mathbf{i}_1$ is the part of $\mathbf{r}_1^P$ perpendicular to $\mathbf{j}_1$.

Coriolis force. $\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = \Omega\,\mathbf{j}_1 \times \left(a\dot\theta\cos\theta\,\mathbf{i}_1 + a\dot\theta\sin\theta\,\mathbf{j}_1\right) = -\Omega a\dot\theta\cos\theta\,\mathbf{k}_1$, so

$$\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = 2m\Omega a\dot{\theta}\cos\theta\,\mathbf{k}_1$$

Total inertia force:

$$\mathbf{F}_i = m\Omega^2 a\sin\theta\,\mathbf{i}_1 + 2m\Omega a\dot{\theta}\cos\theta\,\mathbf{k}_1$$

(c) Weight in $OX_1Y_1Z_1$

The axis of rotation is the vertical, so $\mathbf{j}_1 = \mathbf{j}$ at all times and

$$\mathbf{W} = -mg\,\mathbf{j}_1 \qquad (W_{X_1} = 0,\quad W_{Y_1} = -mg,\quad W_{Z_1} = 0)$$

In the wire basis: $\mathbf{W}\cdot\mathbf{e}_t = -mg\sin\theta$, $\mathbf{W}\cdot\mathbf{e}_n = -mg\cos\theta$, $\mathbf{W}\cdot\mathbf{k}_1 = 0$.

(d) Equilibrium points

> **Pedagogical Justification:** Newton's law in S_1, $m\mathbf{a}_1^P = \mathbf{W} + \mathbf{N} + \mathbf{F}_i$ (Notes Eq. 4.63), projected on the tangent $\mathbf{e}_t$, eliminates the smooth reaction ($\mathbf{N}\cdot\mathbf{e}_t = 0$) and the Coriolis force ($\mathbf{k}_1\cdot\mathbf{e}_t = 0$).

Relative acceleration. Product rule on $a\dot{\theta}\,\mathbf{e}_t$ in S_1, with $\left.\frac{d\mathbf{e}_t}{dt}\right|_1 = \dot{\theta}\,(-\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1) = \dot{\theta}\,\mathbf{e}_n$:

$$\mathbf{a}_1^P = a\ddot{\theta}\,\mathbf{e}_t + a\dot{\theta}^2\,\mathbf{e}_n$$

Tangential equation. $\mathbf{F}_C\cdot\mathbf{e}_t = m\Omega^2 a\sin\theta\cos\theta$, so

$$ma\ddot{\theta} = -mg\sin\theta + m\Omega^2 a\sin\theta\cos\theta \;\Rightarrow\; \ddot{\theta} = \sin\theta\left(\Omega^2\cos\theta - \frac{g}{a}\right)$$

Equilibria (relative rest, $\ddot{\theta} = \dot{\theta} = 0$):

1. $\sin\theta = 0$: $\theta = 0$, the lowest point O. It exists for every Ω.

2. $\cos\theta_* = \dfrac{g}{a\Omega^2}$: two symmetric points $\theta = \pm\theta_*$, which exist only if $\Omega^2 > g/a$ (then $0 < \theta_* < \pi/2$, inside the wire).

Stability with the potential of (f), $V_1(\theta) = mga(1-\cos\theta) - \frac{1}{2}m\Omega^2 a^2 \sin^2\theta$ (the tangential equation is $ma^2\ddot\theta = -dV_1/d\theta$). Its derivatives are $V_1' = ma\sin\theta\left(g - \Omega^2 a\cos\theta\right)$ and $V_1'' = mga\cos\theta - m\Omega^2 a^2\cos 2\theta$:

- $V_1''(0) = ma\left(g - \Omega^2 a\right)$: $\theta = 0$ is stable if $\Omega^2 < g/a$ and unstable if $\Omega^2 > g/a$.
- $V_1''(\pm\theta_*) = mga\cos\theta_* - m\Omega^2 a^2\left(2\cos^2\theta_* - 1\right) = m\Omega^2 a^2\left(1 - \cos^2\theta_*\right) = m\Omega^2 a^2\sin^2\theta_* > 0$ (using $ga = \Omega^2 a^2\cos\theta_*$): the off-axis equilibria are stable whenever they exist.

(e) Quadratures and reaction force

> **Pedagogical Justification:** multiplying the tangential equation by $\dot\theta$ makes both sides exact derivatives; the result is the energy integral of (f). The remaining two projections of Newton's law (on $\mathbf{e}_n$ and $\mathbf{k}_1$) give the reaction.

First integral. Multiply $ma^2\ddot\theta = -mga\sin\theta + m\Omega^2 a^2\sin\theta\cos\theta$ by $\dot\theta$. By the chain rule, $\frac{d}{dt}\left(\frac{1}{2}\dot\theta^2\right) = \dot\theta\ddot\theta$, $\frac{d}{dt}\cos\theta = -\sin\theta\,\dot\theta$ and $\frac{d}{dt}\left(\frac{1}{2}\sin^2\theta\right) = \sin\theta\cos\theta\,\dot\theta$, so

$$\frac{d}{dt}\left[\tfrac{1}{2}ma^2\dot\theta^2 - mga\cos\theta - \tfrac{1}{2}m\Omega^2 a^2\sin^2\theta\right] = 0$$

Integrate from 0 to t and apply Barrow's rule, with $\theta(0) = 0$ and $a\dot\theta(0) = v_0$ (at O, $\mathbf{r}_1^P = \mathbf{0}$, so $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = \mathbf{0}$ and the given absolute velocity equals the relative one, $v_0\,\mathbf{i}_1 = a\dot\theta(0)\,\mathbf{e}_t(0)$):

$$\left[\tfrac{1}{2}ma^2\dot\theta^2 - mga\cos\theta - \tfrac{1}{2}m\Omega^2 a^2\sin^2\theta\right]_0^t = 0 \;\Rightarrow\; \dot\theta^2 = \frac{v_0^2}{a^2} - \frac{2g}{a}\left(1-\cos\theta\right) + \Omega^2\sin^2\theta \equiv \Phi(\theta)$$

Quadrature. Separate variables ($d\theta = \sqrt{\Phi}\,dt$, positive root since $\dot\theta(0) = v_0/a > 0$, valid until the first turning point $\Phi = 0$) and integrate from 0 to t:

$$t = \int_0^{\theta(t)} \frac{d\theta'}{\sqrt{\dfrac{v_0^2}{a^2} - \dfrac{2g}{a}\left(1-\cos\theta'\right) + \Omega^2\sin^2\theta'}}$$

Reaction force. $\mathbf{N} = N_n\,\mathbf{e}_n + N_z\,\mathbf{k}_1$ (no tangential component). Projection on $\mathbf{e}_n$, with $\mathbf{F}_C \cdot \mathbf{e}_n = -m\Omega^2 a \sin^2\theta$:

$$ma\dot\theta^2 = N_n - mg\cos\theta - m\Omega^2 a\sin^2\theta \;\Rightarrow\; N_n = ma\dot\theta^2 + mg\cos\theta + m\Omega^2 a\sin^2\theta$$

Substituting $\dot\theta^2 = \Phi(\theta)$:

$$N_n(\theta) = \frac{mv_0^2}{a} + mg\,(3\cos\theta - 2) + 2m\Omega^2 a\sin^2\theta$$

Projection on $\mathbf{k}_1$ ($\mathbf{a}_1^P \cdot \mathbf{k}_1 = 0$, $\mathbf{W}\cdot\mathbf{k}_1 = 0$):

$$0 = N_z + 2m\Omega a\dot\theta\cos\theta \;\Rightarrow\; N_z = -2m\Omega a\dot\theta\cos\theta = \mp 2m\Omega a\cos\theta\sqrt{\Phi(\theta)}$$

(upper sign while $\dot\theta > 0$). The total reaction is $\mathbf{N} = N_n\,(-\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1) + N_z\,\mathbf{k}_1$.

Staying on the wire. The particle reaches the end $\theta = \pi/2$ only if $\Phi(\pi/2) = v_0^2/a^2 - 2g/a + \Omega^2 \geq 0$; otherwise it oscillates between the two roots of $\Phi(\theta) = 0$.

(f) Energy equation

> **Pedagogical Justification:** E_1^P is conserved when every force that does work in S_1 derives from a time-independent potential (Slide 29; theory note, Sec. 12.6).

Constant Ω: yes. In S_1: the wire is fixed and smooth, so $\mathbf{N}\cdot\mathbf{v}_1^P = 0$; the Coriolis force never does work, $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P = 0$ (Slide 31); the weight has the potential $mgy_1 = mga(1-\cos\theta)$; the centrifugal force has the potential $-\tfrac12 m\Omega^2 r_\perp^2 = -\tfrac12 m\Omega^2 a^2\sin^2\theta$ (Slide 30), time-independent because Ω is constant. Hence

$$E_1^P = \tfrac12 ma^2\dot\theta^2 + mga\,(1-\cos\theta) - \tfrac12 m\Omega^2 a^2\sin^2\theta = \tfrac12 mv_0^2$$

which is the first integral used in (e). (In S_0 the mechanical energy is not conserved: the reaction N_z is perpendicular to the moving wire but not to the absolute velocity, so the motor that keeps Ω constant does work.)

Variable $\Omega(t)$: no energy integral. Two things change. The Euler force appears, $\mathbf{F}_E = -m\dot{\Omega}\,\mathbf{j}_1 \times a\sin\theta\,\mathbf{i}_1 = m\dot{\Omega}a\sin\theta\,\mathbf{k}_1$; it is normal to the plane of the wire, so $\mathbf{F}_E \cdot \mathbf{v}_1^P = 0$ and it does no work in S_1 (it is balanced by N_z). But the centrifugal potential now depends explicitly on time, $V_C = -\frac{1}{2}m\Omega(t)^2 a^2 \sin^2\theta$, and differentiating E_1^P along the motion (tangential equation with $\Omega(t)$) gives

$$\frac{dE_1^P}{dt} = \frac{\partial V_C}{\partial t} = -m\Omega\dot{\Omega}a^2\sin^2\theta \neq 0$$

so E_1^P is not a constant of motion and the problem cannot be reduced to a quadrature with the energy equation. The work-energy relation $dE_1^P/dt = \partial V_C/\partial t$ is still valid, but it is not a first integral.

> **Correction to the Topic 3 solution**
> The Topic 3 version of this problem placed the origin at the centre of the circle (the statement puts it at the lowest point O), gave only one off-axis equilibrium (there are two, $\pm\theta_*$), gave only the out-of-plane reaction (the in-plane component N_n is also part of the answer) and said that for variable Ω "Euler forces do work". The Euler force $m\dot{\Omega}a\sin\theta\,\mathbf{k}_1$ is perpendicular to $\mathbf{v}_1^P$ and does no work; the energy integral is lost because the centrifugal potential depends explicitly on time. The quadrature $\dot{\theta}^2 = \Phi(\theta)$, the equilibrium $\cos\theta_* = g/(a\Omega^2)$ and $N_z = -2m\Omega a\dot{\theta}\cos\theta$ of the Topic 3 version agree with this solution.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Bifurcation.** For $\Omega^2 < g/a$ gravity dominates and the bottom is the only (stable) equilibrium. At $\Omega_c = \sqrt{g/a}$ the bottom becomes unstable and two stable equilibria $\pm\theta_*$ appear, where the tangential components of weight and centrifugal force balance: $mg\sin\theta_* = m\Omega^2 a\sin\theta_*\cos\theta_*$.
- **Limits.** $\Omega \to 0$: $\Phi = v_0^2/a^2 - 2g(1-\cos\theta)/a$ and $N_n = mv_0^2/a + mg(3\cos\theta - 2)$, the simple-pendulum results; $N_z \to 0$. $\Omega \to \infty$: $\cos\theta_* \to 0$, the equilibria move to the ends of the wire.

- **Coriolis and the reaction.** N_z is exactly the force the wire must exert to cancel the Coriolis force; it changes sign with $\dot\theta$, pushing the particle forward or backward with the rotation.
- **Dimensions:** $[\Omega^2] = [g/a] = \mathrm{s}^{-2}$, so every term of Φ is in s^{-2} and $[t] = \mathrm{rad}/\mathrm{s}^{-1} = \mathrm{s}$; $[mv_0^2/a] = [mg] = [m\Omega^2 a] = \mathrm{kg}\cdot\mathrm{m}\cdot\mathrm{s}^{-2} = \mathrm{N}$; $[m\Omega a\dot\theta] = \mathrm{N}$; $[E_1^P] = [ma^2\dot\theta^2] = \mathrm{J}$.
- **Symbolic check (sympy, `verify_P26.py`):** $[{}_0R_1]$ and $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}$; Notes Eq. 3.27 against the double derivative of $\mathbf{r}_1^P$ in $\mathcal{B}_0$; $\mathbf{F}_C, \mathbf{F}_{COR}$; the equation of motion and both reaction components from $m\mathbf{a}_0^P = \mathbf{W} + \mathbf{N}$ in S_0; $dE_1^P/dt = 0$; $\Phi(\theta)$ and $N_n(\theta)$; the stability derivatives; and, for $\Omega(t)$, the Euler force, its zero power and $dE_1^P/dt = -m\Omega\dot\Omega a^2\sin^2\theta$. A numerical integration ($g = 9.81$, $a = 1$, $\Omega = 4$, $v_0 = 2$, SI) keeps $\dot\theta^2 - \Phi(\theta)$ below 3×10^{-10}.

Source: Problems.pdf (PDF page 14)